

Elisabetta Bettini

Mechanical Engineer — Mechanism Design & Structural Dynamics — FEM & Experimental Analysis

Pasadena, CA, USA

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Summary

Mechanical Engineer specializing in structural dynamics, mechanism design, and high-fidelity FEM modeling, currently at NASA Jet Propulsion Laboratory. Background includes design, kinematic and singularity analysis of multi-DoF linkages, hyperelastic FEM modeling of soft robotic components, and actuator sizing via simulation under worst-case loading conditions. Experienced in mechanical prototyping and hardware integration, including sensors and actuators, as well as modal analysis and FEM-experimental correlation for precision mechanical systems.

Professional Experience

NASA Jet Propulsion Laboratory (JPL)

Pasadena, CA

Structural Dynamics Intern - Robotics Mobility and Manipulation

Nov 2025 – Present

- Developed high-fidelity FEM models of a segmented telescope testbed for dynamic characterization
- Performed modal analysis, extracting natural frequencies and mode shapes
- Supported experimental modal testing
- Conducted FEM-experimental correlation to validate and refine structural models
- Developed a steady-state lumped thermal model of the telescope using a node-based conductive-convective network

Piaggio & C. S.p.A.

Pontedera, IT

Aftersales Engineer - Training & Technical Documentation

Aug 2023 – Oct 2025

- Produced and managed technical documentation for diagnostic procedures of electromechanical ECU systems
- Collaborated with R&D teams to improve diagnostic accuracy and fault detection strategies
- Supervised publication of technical notes and communications

PERCeptual Robotics Lab – Scuola Superiore Sant’Anna

Pisa, IT

Master Thesis Researcher

Dec 2022 – Jun 2023

- Designed a master haptic interface for remote ultrasound diagnostics, featuring a five-bar mechanism with a spherical joint for precise probe manipulation
- Performed kinematic and singularity analysis to optimize workspace and motion fidelity
- Used MATLAB to perform singularity analysis and actuator sizing based on kinetostatic modeling
- Designed and prototyped mechanical components using PTC Creo and additive manufacturing

3D Printing and Rapid Prototyping Lab – University of Pisa

Pisa, IT

Bachelor Thesis Researcher

Mar 2019 – Feb 2020

- Investigated surface roughness in stereolithography 3D-printed photopolymers
- Analyzed effects of layer height, orientation, and resin type
- Conducted pre- and post-curing surface characterization

Education

Visiting Graduate Research Student

2025 – Present

Aerospace Engineering

California Institute of Technology - CALTECH

Postgraduate Master (II Level) - SPICES

2025 – Present

Space Missions Science, Design and Applications

University of Bologna GPA: 3.8

M.Sc. Mechanical Engineering

2020 – 2023

University of Pisa GPA: 3.7

Exchange Program – University of Illinois Urbana-Champaign

2022

Energy Systems (NPPE 201)

Technical Skills & Software

Simulation & Analysis: Abaqus, Ansys Workbench, Ansys APDL, MATLAB/Simulink, Mathcad

Mechanical Design: SolidWorks, PTC Creo, Autodesk Inventor

Graphics & Visualization: Adobe Illustrator, Adobe InDesign, Autodesk SketchBook Pro

Audio/Video: Melodyne, Adobe Premiere Pro, Adobe After Effects, Movavi Video Editor Plus

Other: Microsoft Office Suite

Languages

Italian (native), English (C1 - Cambridge CAE), French (B1)

Leadership & Volunteering

- Rotaract Club Pontedera – Councilor (2024–2025), Prefect (2023–2024), Secretary (2022–2023)
- Member, CISOM – Corpo Italiano di Soccorso dell’Ordine di Malta (2019–2025)
- RYLA – Rotary Youth Leadership Awards (2022)

Hobbies & Activities

Digital art, graphic design, and singing.

Music composition and video production.